

AI VELOCITY TRADING

Autonomous SPY Weekly Iron Butterfly Engine

Alpaca AI Trading Agents Hackathon Submission | Account Portfolio: \$100,000 USD

Addressing Options Complexity & Tail Risk

01. Manual Execution Risk

Multi-leg option spreads require executing 4 atomic legs simultaneously. Manual execution leads to severe leg-out risk, latency, and slippage.

02. Regime Blindness

Static options selling strategies suffer catastrophic losses during sudden volatility expansions or strong macro directional trends.

03. Assignment & Pin Risk

Holding short weekly options contracts into expiration exposes portfolios to weekend gap risk and assignment uncertainty.

AI Market-Regime Classifier & Strike Engine

1. Ensemble Regime Classifier

- Predicts 5-day mean-reversion probability using Random Forest & XGBoost.
- Features: VIX levels, SPY realized volatility, moving average dispersion.
- Hard Execution Gate: Only deploys when range-bound confidence exceeds 70%.

2. Dynamic Strike Selection Engine

- ATM Center Legs: Sells nearest 0.50 Delta call and put at current spot.
- OTM Protective Wings: Dynamically anchors long wings at 1.0σ IV expected move.
- IV Skew Filter: Confirms options premium is sufficiently rich prior to deployment.

Multi-Layered Automated Risk Gates

Macro Event Guard [Pre-Trade Gate]

Programmatically halts execution during high-impact scheduled economic events (FOMC, CPI, NFP).

Credit Multiplier [Pre-Trade Gate]

Enforces minimum net credit received $\geq 25\%$ of total wing width to guarantee positive risk/reward.

Max Loss Circuit Breaker [In-Trade Gate]

Automated stop-loss triggers immediately if basket drawdown reaches 100% of net credit.

Pin Risk Exit Gate [In-Trade Gate]

Unconditionally force-closes all legs 2 hours prior to Friday market close to eliminate assignment.

Alpaca API & Infrastructure Implementation

```
# Alpaca Atomic Multileg Execution
order_data = OrderRequest(
    symbol="SPY",
    qty=1,
    time_in_force=TimeInForce.DAY,
    order_class=OrderClass.MULTILEG,
    legs=[
        {"symbol": long_call, "side": OrderSide.BUY},
        {"symbol": short_call, "side": OrderSide.SELL},
        {"symbol": short_put, "side": OrderSide.SELL},
        {"symbol": long_put, "side": OrderSide.BUY}
    ]
)
trading_client.submit_order(order_data)
```

Key Infrastructure Capabilities

- Atomic Multi-Leg Routing: Eliminates leg-out execution risk by placing all 4 option legs in a single order submission.
- Alpaca MCP Integration: Model Context Protocol (MCP) server integration for context-aware model querying.
- Real-time Streaming: WebSocket connection for instant order fills and continuous option basket re-valuation.

Live Paper Trading Dashboard Highlights

\$100,000

Starting Paper Capital

Dedicated fresh hackathon environment

Sub-Second

Execution Latency

Atomic 4-leg placement via alpaca-py

45% - 50%

Profit Capture Target

Automated profit-taking to recycle capital

100%

Defined Max Risk

Capped strictly at wing width minus net credit

AI Velocity Trading

- ✓ **Systematic Market-Neutral Yield:** Captures theta decay on weekly SPY options.
- ✓ **Predictive Intelligence:** Machine Learning regime filter prevents trading into volatile trends.
- ✓ **Programmatic Capital Protection:** Isolated pre-trade and in-trade safety gates eliminate tail risk.
- ✓ **Production Ready:** Built natively on Alpaca Options API & Alpaca MCP Server.